

Prevention of congenital abnormalities by periconceptional multivitamin supplementation //

Andrew E Czeizel

Abstract

Objective—To study the effect of periconceptional multivitamin supplementation on neural tube defects and other congenital abnormality entities.

Design—Randomised controlled trial of supplementation with multivitamins and trace elements.

Setting—Hungarian family planning programme.

Subjects—4156 pregnancies with known outcome and 3713 infants evaluated in the eighth month of life.

Interventions—A single tablet of a multivitamin including 0.8 mg of folic acid or trace elements supplement daily for at least one month before conception and at least two months after conception.

Main outcome measures—Number of major and mild congenital abnormalities.

Results—The rate of all major congenital abnormalities was significantly lower in the group given vitamins than in the group given trace elements and this difference cannot be explained totally by the significant reduction of neural tube defects. The rate of major congenital abnormalities other than neural tube defects and genetic syndromes was 9.0/1000 in pregnancies with known outcome in the vitamin group and 16.6/1000 in the trace element group; relative risk 1.85 (95% confidence interval 1.02 to 3.38); difference, 7.6/1000. The rate of all major congenital abnormalities other than neural tube defects and genetic syndromes diagnosed up to the eighth month of life was 14.7/1000 in the vitamin group and 28.3/1000 in the trace element group; relative risk 1.95 (1.23 to 3.09); difference, 13.6/1000. The rate of some congenital abnormalities was lower in the vitamin group than in the trace element group but the differences for each group of abnormalities were not significant.

Conclusions—Periconceptional multivitamin supplementation can reduce not only the rate of neural tube defects but also the rate of other major non-genetic syndromic congenital abnormalities. Further studies are needed to differentiate the chance effect and vitamin dependent effect.

Introduction

Periconceptional supplementation with a multivitamin including folic acid^{1,2} or with folic acid alone^{1,4} can reduce the recurrence of neural tube defects. The Hungarian randomised controlled trial showed that the occurrence of neural tube defects could be appreciably reduced by giving periconceptional multivitamin supplements (including 0.8 mg of folic acid).⁵ This primary preventive method has great public health significance, being a better alternative than prenatal diagnosis and selective abortion. In addition to the lessening of human suffering, supplementation would decrease medical costs. The expected extensive use of periconceptional multivitamin and folic acid supplementation demands studies on the effects of this preventive method on pregnancy outcomes and maternal health.⁶ The Hungarian randomised controlled trial was appropriate to study the effect of pericon-

ceptional multivitamin supplementation on congenital abnormalities other than neural tube defects, which is the subject of this paper.

Methods

The Hungarian randomised controlled trial of periconceptional multivitamin supplementation was part of the Hungarian family planning programme, which was established on 1 February 1984. Couples who satisfy three criteria are eligible for participation in this programme: no delayed conception or infertility (that is, inability to conceive after more than 12 months of sexual activity without contraception); not currently pregnant; voluntary participation. In the first four years of the programme, there were two other criteria: age of the woman under 35 (older women were referred to the genetic counselling clinic) and no previous wanted pregnancy (induced abortion was not a reason for exclusion). The programme involves three main steps performed or supervised by qualified nurses: a check of reproductive health; a three month preparation for conception including multivitamin or trace element supplementation; and protection of early pregnancy.⁶

At the first visit eligible couples were informed about the purpose of multivitamin use and about the "blind" use of one of two kinds of tablets—that is, they were asked whether they agreed to be allocated to either group on the basis of a randomisation table. After this, the women were asked to sign a written informed consent and to stop taking other vitamins. The participants were advised to take a single tablet of either the "vitamin," Elevit Pronatal (Roche), composed of 12 vitamins (vitamin A 6000 IU till the end of 1989 and 4000 IU in 1990-1, B-1 1.6 mg, B-2 1.8 mg, nicotinamide 19.0 mg, B-6 2.6 mg, calcium pantothenate 10 mg, biotin 0.2 mg, B-12 4.0 µg, C 100 mg, D 500 IU, E 15 mg, folic acid 0.8 mg), four minerals (calcium 125 mg, phosphorous 125 mg, magnesium 100 mg, iron 60 mg), and three trace elements (copper 1 mg, manganese 1 mg, zinc 7.5 mg) or the "trace element" supplement (copper 1 mg, manganese 1 mg, zinc 7.5 mg, and vitamin C 7.5 mg) each day for one month before planned conception.

At the second visit women were supplied with tablets for a further three months and were asked to attempt to conceive and to visit us again immediately after the first missed menstrual period. If the participant did not conceive within three months she was supplied with tablets for the next three months, a month at a time. The purpose of the third visit was the confirmation of pregnancy by a sensitive serum pregnancy test after the first missed menstrual period. Pregnancy was also confirmed by ultrasonography within two weeks. Further tablets were supplied until the 12th week of gestation. The fourth visit was a farewell meeting at the 12th week of pregnancy because after this the pregnant women were referred to other clinics for routine antenatal care. The compliance with the regimen of supplementation was verified verbally in discussion with the women; by evaluating the check marks on the form for the basal body

Department of Human Genetics and Teratology, National Institute of Hygiene—WHO Collaborating Centre for the Community Control of Hereditary Diseases, Budapest, Hungary
Andrew E Czeizel, medical geneticist

Correspondence to: Dr Czeizel, OKI 1966 Budapest, Gyűjtőút 2-6, Hungary.

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temperature measurement; and by counting unused tablets when boxes were returned. The criteria for full, partial, and no supplementation have been presented elsewhere.⁵

After the end of the pregnancy we were sent the completed certificate for each of the participants which included the date of pregnancy outcome and any congenital abnormalities of prenatally diagnosed and stillborn fetuses or liveborn infants. The certificate was filled in by the mothers and confirmed by their physician. In Hungary all deliveries and terminations of pregnancies take place in obstetrical inpatient clinics. Considerable effort was undertaken to assure that pregnancy outcomes of all women with a confirmed pregnancy were evaluated. If the completed certificate was not received one month after the expected date of delivery, a letter was mailed to the women asking her to send the completed certificate or discharge summary, or both. If there was no response a member of staff visited her at home. Only after such efforts failed were the participants considered to have dropped out from the study.

Pregnancies with known outcomes¹ (prenatally diagnosed and affected fetuses of pregnancies terminated in the second trimester; stillborn fetuses and liveborn infants known to have, or be free of, congenital abnormalities) were evaluated with particular attention to the occurrence and correct diagnosis or description of congenital abnormalities. After the report of a case as having congenital abnormality, as much clinical and pathological information as possible was obtained, including any prenatal ultrasonography films, a

detailed physician's description (discharge summary, a written report from the examining physician, etc), and necropsy records. The data concerning congenital abnormalities were evaluated by an independent data monitoring group every six months.

Infants were given a physical examination during the eighth month of life after a mail or telephone invitation to attend the family planning centre. After review of medical documents concerning previously diagnosed congenital abnormalities and diseases, treatments, operations, the examination included a paediatric evaluation performed "blind" by two paediatricians. If families did not take part in these examinations, they were invited again to participate. If there was still no response we contacted the infant's paediatrician and the medical history, particularly data concerning congenital abnormalities, diagnosed after birth, were obtained.

Two-tailed χ^2 and Fisher exact tests were used for statistical evaluation.

Results

The code was broken at the end of 1991. Of 4753 participants with confirmed pregnancies, pregnancy outcomes were ascertained in 4704 cases (drop out 1%). The numbers of women given full, partially, and no supplements (1699+471+224 v 1682+451+177) were not significantly different between vitamin and trace element groups. Demographic factors were similar in the two groups (table I). Congenital abnormalities were divided into three groups: lethal and severe (which together constitute major abnormalities) and mild.⁹ Minor anomalies were excluded from this analysis.

Table II shows the distribution of cases with major congenital abnormalities. There were six cases of neural tube defects in the group receiving the trace element supplement compared with none in the vitamin group.⁵ Isolated hydrocephalus did not occur in the vitamin group but two cases with congenital hydrocephalus were found in the trace element group. Some experts (for example, Lorber,¹⁰ have suggested that some cases of isolated hydrocephalus may properly be regarded as part of the neural tube defect spectrum. Four newborn infants with cleft lip with or without cleft palate were found in the vitamin group and three in the trace element group. (One case in the

TABLE I—Characteristics of study population and outcome of pregnancies

	Group given vitamins	Group given trace elements
No of participants with confirmed pregnancy	2420	2333
No (%) of dropouts	26 (1.1)	23 (1.0)
No of evaluated pregnancies	2394	2310
Mean (SD) maternal age (years)	26.8±3.4	26.7±3.3
No (%) primiparous women	94.5	94.0
No of pregnancies with known outcome	2104	2052
Termination after prenatal diagnosis of defects	3	13
Late fetal death (stillbirth)	11	7
Live birth	2090	2032
No (%) of infants evaluated during eighth month of life	1876 (89.8)	1837 (90.4)

TABLE II—Number of major congenital abnormality entities in groups studied

Congenital abnormality entities	Group given vitamins			Group given trace elements		
	Pregnancies with known outcome (n=2104)	Infants assessed during 8th month of life (n=1876)	Total	Pregnancies with known outcome (n=2052)	Infants assessed during 8th month of life (n=1837)	Total
Neural tube defects	0	0	0	6	0	6
Genetic syndromes						
Down's syndrome	2	0	2	3	0	3
Allagile syndrome, collodion baby	2	0	2	0	0	0
Congenital hydrocephalus	0	0	0	2	0	2
Buphthalmos	0	2	2	0	0	0
Cardiovascular malformations	6	4	10	9	8	17
Cleft palate	0	0	0	2	0	2
Cleft lip with or without cleft palate	4	0	4	3	0	3
Congenital hypertrophic pyloric stenosis	0	1	1	0	4	4
Hypospadias	1	1	2	1	2	3
Genital anomalies of male*	0	1	1	0	3	3
Obstructive defects of urinary system	1	0	1	2	1	3
Renal agenesis (Potter sequence)	0	0	0	1	1	2
Congenital limb deficiency	1	0	1	5	0	5
Exomphalos	1	0	1	1	0	1
Other isolated abnormalities	2†	1‡	3	5	3§	8
Multiple congenital abnormalities	3	2	5	3	2	5
Total	23	12	35	43	24	67
Total excluding neural tube defects and genetic syndromes	19	12	31	34	24	58

*At least two of: hypospadias, undescended testis, congenital inguinal hernia.

†Tracheal stenosis; cystic kidney type II.

‡Cerebellar agenesis.

||Microcephalus; non-immune hydrops; anal atresia; extrophy of urinary bladder; arthrogryposis.

§Congenital cerebellar cyst; syndactyly; craniosynostosis.

TABLE III—Distribution of cardiovascular malformations in groups studied

	Group given vitamins			Group given trace elements		
	Pregnancies with known outcome (n=2104)	Infants assessed during 8th month of life (n=1876)	Total	Pregnancies with known outcome (n=2052)	Infants assessed during 8th month of life (n=1837)	Total
Cardiovascular malformations						
Transposition of great vessels	0	0	0	1	0	1
Tetralogy of Fallot	0	0	0	1	0	1
Ventricular septal defect	0	2	2	3	5	8
Atrial septal defect, type II	1	0	1	1	2	3
Congenital aortic stenosis	1	2	3	2	0	2
Hypoplastic left heart	2	0	2	1	0	1
Patent ductus arteriosus	0	0	0	0	1	1
Other cardiovascular malformations	2*	0	2	0	0	0
Total	6	4	10	9	8	17

*Atrial septal defect, type I and ventricular septal defect; double onset of pulmonary artery with pseudovalves and stenosis.

TABLE IV—Number of mild congenital abnormality entities in groups studied

	Group given vitamins			Group given trace elements		
	Pregnancies with known outcome (n=2104)	Infants assessed during 8th month of life (n=1876)	Total	Pregnancies with known outcome (n=2052)	Infants assessed during 8th month of life (n=1837)	Total
Congenital abnormalities entities						
Undescended testicles	0	7	7	0	5	5
Congenital dislocation of the hip	0	28	28	0	18	18
Deformities of feet	0	10	10	1	13	14
Congenital postural deformity association*	2	2	4	0	4	4
Pectus excavatum	0	1	1	0	0	0
Foramina parietale permagna	0	0	0	2	0	2
Lange haemangioma	3	0	3	1	0	1
Dermoid cyst	0	0	0	0	1	1
Congenital inguinal hernia	0	22	22	0	25	25
Total	5	70	75	4	66	70

*At least two of: congenital dislocation of hip, deformities of feet, torticollis and other deformities.

vitamin group was born to an epileptic woman treated with valproate.) This finding is not in agreement with the report of Tolarova,¹¹ who found that periconceptional multivitamin supplementation with a dose of 10 mg folic acid per day protected against recurrence of oral cleft in subsequent children. Two children with cleft palate were found in the trace element group and none in the vitamin group.

Some other congenital abnormality entities had a lower rate in the vitamin group than in the trace element group but the difference was not significant for individual abnormalities. However, a general trend is worth mentioning. The intention to treat analysis (when fully, partially, and unsupplemented cases were evaluated together in the vitamin or trace element groups) showed that the total rate of cases with major congenital abnormalities in informative pregnancies whose outcome was known was significantly higher in the trace element group than in the vitamin group (21.0/1000 *v* 10.9/1000; $\chi^2_1=6.68$, $p=0.01$). When neural tube defects were excluded the difference was not significant ($\chi^2_1=3.68$; $p=0.055$). It seemed reasonable to exclude genetic syndromes (Down, Alagille, collodion baby) as well, and then, the number of cases was 19 (9.0/1000) in the vitamin group and 34 (16.6/1000) in the trace element group; the difference was significant ($\chi^2_1=4.69$; $p=0.03$) and the relative risk was 1.85 (95% confidence interval 1.02 to 3.38). Multiple abnormalities (two or more different abnormalities in the same fetus or infant¹²) were found in 10 cases that could not be classified as having congenital abnormality syndrome.

The number of major congenital abnormalities in infants evaluated during the eighth month of life was significantly higher in the trace element group than in the vitamin group ($\chi^2_1=4.30$; $p=0.038$).

The total rate of all major congenital abnormalities diagnosed from the second trimester of gestation until the eighth month of life was significantly different between the vitamin and trace element groups ($\chi^2_1=11.13$; $p=0.0008$). After cases with neural tube defects

and genetic syndromes were excluded there were 31 cases in the vitamin group and 58 in the trace element group, rates of 14.7 and 28.3, respectively, per 1000 pregnancies of known outcome. The difference was highly significant ($\chi^2_1=9.08$; $p=0.0025$) and the relative risk was 1.95 (1.23 to 3.09).

The differences in the congenital abnormality rates between the vitamin and trace element groups were explained for the most part by congenital limb deficiencies (1 in the vitamin group *v* 5 in the trace element group; $p=0.124$, Fisher's exact test), defects of the urinary system—that is, renal agenesis and obstructive defects (1 *v* 5; $p=0.124$), congenital hypertrophic pyloric stenosis (1 *v* 4; $p=0.21$), and cardiovascular malformations (10 *v* 17; $p=0.16$) (table III). The difference was close to significance in ventricular septal defect (8 *v* 2; $p=0.06$), but the diagnosis was based only on clinical findings in eight cases. Until the eighth month of life only one infant in the trace element group had surgery, and ventricular septal defect was found at necropsy of a fetus with a birth weight of 770 g in the vitamin group.

The number of cases with mild congenital abnormalities was not significantly different between the vitamin and trace element groups in pregnancies with a known outcome ($\chi^2_1=0.38$; $p=0.54$) and in infants assessed during the eighth month of life ($\chi^2_1=0.05$; $p=0.82$) (table IV). Some mild congenital abnormalities were present in surprisingly high numbers (for example, deformities of feet), but the predetermined diagnostic criteria for mild congenital abnormalities (for example, surgery to correct congenital inguinal hernia; Pavlik pillow for at least three months in congenital dislocation of the hip; orthopaedic treatment, mainly plaster casts, in deformities of feet; undescended testis after the third month of life) were followed at the eight month examination.

All congenital dislocations of the hip were diagnosed through neonatal orthopaedic screening; thus these cases had only an Ortolan click—that is, a predisposition to this congenital abnormality. The difference

between the groups ran in the opposite direction to most other congenital abnormalities and was not significant ($\chi^2_1=1.99$; $p=0.157$).

Discussion

Congenital abnormalities diagnosed from the second trimester of pregnancy until the age of 1 year are expected to occur in 65/100 births in Hungary.⁹ The observed rate of congenital abnormalities (67/1000) was close to this figure in the trace element group.

The observed rate of cases with major congenital abnormalities other than neural tube defects and genetic syndromes diagnosed prenatally or immediately after birth was significantly lower in the group given vitamins than in the group given trace elements. The difference between these two groups, 7.6 per 1000 in pregnancies whose outcome was known, is 2.7 times higher than the prevalence of neural tube defects at birth (2.8/1000) in Hungary.¹³ The rate of all cases with major congenital abnormalities diagnosed till the eighth month of life, other than genetic syndromes and neural tube defects, was also significantly lower in the vitamin group than in the trace element group. The difference between two groups, 13.6 per 1000, is more than 4.9 times the rate of neural tube defects. Thus, periconceptual supplementation with multivitamins including folic acid seems to prevent congenital abnormalities.

The incidence of mild congenital abnormalities was not significantly different between the vitamin and trace element groups. The high rate of congenital dislocation of the hip in Hungary is well known.^{14,15}

VITAMINS AND AETIOLOGY OF CONGENITAL ABNORMALITIES

Congenital abnormality entities have specific genetic or environmental teratogenic aetiologies, or both; thus it is difficult to suppose that periconceptual multivitamin supplementation could reduce the occurrence of more than one congenital abnormality entity (that is, neural tube defect). However, some biological compounds (such as antioxidants and aspirin) are known to have different effects, and vitamins may have a similar role. Foliates are essential as coenzymes for the synthesis of both purines and pyrimidines; they are involved in nucleic acid synthesis and are therefore essential to all cell division.¹⁶ Cell division is exceptionally rapid at critical stages of specific developmental fields in fetuses. Deficiency of folic acid was shown to be a risk factor for multiple defects,¹⁷ cardiovascular malformations,^{18,19} and congenital abnormalities of the urinary system²⁰ in experimental studies. The teratogenic effect of folic acid deficiency due to folic acid antagonists such as aminopterin and methotrexate caused, among other defects, limb deficiency in human embryos.^{21,22} Folic acid may not be the only active compound preventing the diverse congenital abnormalities reported here. Folic acid, vitamin B-6, vitamin B-12, vitamin C, and zinc interact with one another in many metabolic pathways. In addition, vitamin dependencies rather than vitamin deficiencies may have some causal role in the origin of neural tube defects²³ and some other congenital abnormalities.

Periconceptual multivitamin supplementation can reduce the occurrence of neural tube defects and the results of this study also indicate a significant decrease in the rate of other major congenital abnormalities. Further studies are needed to differentiate the chance

effect and effect of vitamins in the origin of these congenital abnormalities and to clarify the mechanism of this primary preventive method.

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Correction

Cholesterol lowering and mortality: the importance of considering initial level of risk

Three authors' errors occurred in this paper by George Davey Smith and others (22 May, pp 1367-73). The trial by Ornish (reference 32a, table I) was incorrectly included, since stress management and exercise, as well as cholesterol lowering diet, were prescribed to the treatment group and it was therefore not a single factor trial. Since only one death occurred in this small trial the exclusion of this trial has no influence on the reported results. In the table of trials (table I) niacin is incorrectly spelt as niacine. In the discussion the follow up period for the Renfrew and Paisley survey is given as 12 years; this should read 15 years.